

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 4: Processor Fundamentals

Subtopic 4.2: Assembly Language

Concept 4.2.5: Modes of Addressing

Total Questions: 11

Mark Schemes begin on Page 18

Question 1 | Source: 9618_w25_qp_12 (Q3.a.i)

3 (a) (i) State what is meant by relative addressing.

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..... [1]

Question 3 | Source: 9618_s25_qp_12 (Q7.a.ii)

- 7 (a) The following table shows part of the instruction set for a processor. The processor has one register: the Accumulator (ACC).

Instruction		Explanation
Opcode	Operand	
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDM	#n	Immediate addressing. Load the number n to ACC
STO	<address>	Store the contents of ACC at the given address
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC)
CMP	<address>	Compare the contents of ACC with the contents of <address>
CMP	#n	Compare the contents of ACC with number n
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JMP	<address>	Jump to the given address
END		Return control to the operating system

ACC denotes Accumulator
 <address> can be an absolute or a symbolic address
 # denotes a denary number, e.g. #123
 B denotes a binary number, e.g. B01001010
 & denotes a hexadecimal number, e.g. &4A

The current contents of memory are:

Address Instruction

10	12
11	11
12	10
13	22
14	22
...	
100	LDD 10
101	ADD 12
102	STO 11
103	CMP 14
104	JPE 107
105	INC ACC
106	JMP 101
107	STO 10
108	END

(ii) State the effect of changing instruction `LDD 10` in address 100 to `LDM #10`

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..... [1]

Question 4 | Source: 9618_s25_qp_12 (Q7.a.iii)

- 7 (a) The following table shows part of the instruction set for a processor. The processor has one register: the Accumulator (ACC).

Instruction		Explanation
Opcode	Operand	
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDM	#n	Immediate addressing. Load the number n to ACC
STO	<address>	Store the contents of ACC at the given address
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC)
CMP	<address>	Compare the contents of ACC with the contents of <address>
CMP	#n	Compare the contents of ACC with number n
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JMP	<address>	Jump to the given address
END		Return control to the operating system

ACC denotes Accumulator
 <address> can be an absolute or a symbolic address
 # denotes a denary number, e.g. #123
 B denotes a binary number, e.g. B01001010
 & denotes a hexadecimal number, e.g. &4A

The current contents of memory are:

Address Instruction

10	12
11	11
12	10
13	22
14	22
...	
100	LDD 10
101	ADD 12
102	STO 11
103	CMP 14
104	JPE 107
105	INC ACC
106	JMP 101
107	STO 10
108	END

- (iii) Identify **and** describe **one** mode of addressing **not** given in the table of instructions in part (a).

Mode of addressing

Description

.....

.....

[2]

Question 5 | Source: 9618_s25_qp_13 (Q5.a.ii)

- 5 The following table shows part of the instruction set for a processor. The processor has two registers: the Accumulator (ACC) and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
STO	<address>	Store the contents of ACC at the given address
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
SUB	<address>	Subtract the contents of the given address from the ACC
SUB	#n/Bn/&n	Subtract the number n from the ACC
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
JMP	<address>	Jump to the given address
CMP	<address>	Compare the contents of ACC with the contents of <address>
CMP	#n	Compare the contents of ACC with number n
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
END		Return control to the operating system
ACC denotes Accumulator <address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

(a) The current contents of memory are:

Address	Instruction
50	47
51	48
52	49
53	50
...	
500	LDM #50
501	DEC ACC
502	CMP 51
503	JPE 505
504	JMP 501
505	STO 50
506	SUB #10
507	STO 51
508	END

(ii) Complete the table by identifying **and** describing **two** modes of addressing that are **not** used in the program in part (a).

Mode of addressing	Description
.....	
.....	

[4]

Question 6 | Source: 9618_w24_qp_12 (Q8.a)

- 8 The following table shows part of the instruction set for a processor. The processor has two registers, the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
SUB	#n/Bn/&n	Subtract the number n from the ACC
SUB	<address>	Subtract the contents of the given address from the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

(a) The current contents of memory are shown:

Address	Data
19	25
20	23
21	2
22	4
23	15
24	50
25	22

The current contents of the ACC and IX are shown:

ACC	50
IX	20

Complete the table by writing the content of the ACC and the IX after each set of instructions has run.

	Instructions	ACC content	IX content
1	LDM #19 DEC ACC		
2	LDD 23 ADD 19		
3	LDI 25 INC ACC		
4	LDR #21 LDX 2		

[5]

Question 7 | Source: 9618_w24_qp_13 (Q7.a)

- 7 The following table shows part of the instruction set for a processor. The processor has two registers, the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
SUB	#n/Bn/&n	Subtract the number n from the ACC
SUB	<address>	Subtract the contents of the given address from the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

(a) The current contents of memory are shown:

Address	Data
50	54
51	55
52	50
53	52
54	100
55	25
56	50

The current contents of the ACC and IX are shown:

ACC	50
IX	45

Complete the table by writing the content of the ACC after each program has run.

	Instructions	ACC content
1	LDD 50 ADD #4 ADD 54	
2	LDI 53 DEC ACC ADD 56	
3	LDM #55 SUB #5	

[3]

Question 8 | Source: 9618_w23_qp_11 (Q8.b.ii)

(b) The following table shows part of the instruction set for a processor. The processor has two registers, the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDR	#n	Immediate addressing. Load the number n to IX
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	#n	Compare the contents of ACC with number n
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
OUT		Output to the screen the character whose ASCII value is stored in ACC
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123		

(ii) The instruction `LDR #2` uses immediate addressing.

Give **one** similarity and **one** difference between direct addressing and indexed addressing.

Similarity

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.....

Difference

.....

.....

[2]

Question 9 | Source: 9618_w23_qp_11 (Q8.b.iii)

(b) The following table shows part of the instruction set for a processor. The processor has two registers, the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDR	#n	Immediate addressing. Load the number n to IX
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	#n	Compare the contents of ACC with number n
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
OUT		Output to the screen the character whose ASCII value is stored in ACC
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123		

(iii) Identify **one other** mode of addressing.

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..... [1]

Question 10 | Source: 9618_s23_qp_12 (Q3.b)

3 A program is written in assembly language.

(b) Assembly language statements can use different modes of addressing.

Complete the following description of addressing modes.

..... addressing is when the operand holds the memory address of the data.

..... addressing is when the operand holds a memory address that stores the memory address of the data.

..... addressing is when the operand is the data.

[3]

Question 11 | Source: 9618_w22_qp_13 (Q6.a.ii)

- 6 The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
MOV	<register>	Moves the contents of the accumulator to the given register (IX)
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
ADD	#n	Add the denary number n to the ACC
SUB	#n	Subtract the denary number n from the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
JMP	<address>	Jump to the given address
CMP	#n	Compare the contents of ACC with number n
CMI	<address>	Indirect addressing. The address to be used is at the given address. Compare the contents of ACC with the contents of this second address
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

(a) The current contents of main memory and selected values from the ASCII character set are given.

(ii) Explain the purpose of **relative addressing** in an assembly language program.

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..... [2]

Question 12 | Source: 9618_w21_qp_12 (Q8.b.iii)

8 The Von Neumann model for a computer system uses registers.

- (b) The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
STO	<address>	Store the contents of ACC at the given address
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
IN		Key in a character and store its ASCII value in ACC
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left hand end
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123		

The current contents of main memory are shown:

Address	Data
100	01010101
101	11110000
102	00001111
103	00000000
104	11111111

(iii) The opcode `LDM` uses immediate addressing. The opcode `LDD` uses direct addressing.

Identify **and** describe **one additional** mode of addressing.

Mode of addressing

Description

.....

.....

[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q3.a.i)

3(a)(i)	1 mark The value of the operand is an offset value which is added to another base value to give the address from which the contents are loaded to the accumulator	1
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Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q6.c)

6(c)	1 mark for each bullet point, max 3 marks - Data movement - Arithmetic operations - Conditional and unconditional (jump) instructions	3
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Mark Scheme for Question 3 | Source: 9618_s25_ms_12 (Q7.a.ii)

7(a)(ii)	1 mark from: - The number 10 will be loaded into the ACC instead of 12 / contents of address 10 - The addition will give 20 not 22 so the comparison will fail and result in an infinite loop	1
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Mark Scheme for Question 4 | Source: 9618_s25_ms_12 (Q7.a.iii)

7(a)(iii)	1 mark for name of mode, 1 mark for matching description e.g. - Indirect - The operand points to the memory location that contains the address of the data - Indexed - The address of the data is formed by adding the contents of the Index Register (IX) to the operand - Relative - The address is calculated using its distance from a base address	2
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Mark Scheme for Question 5 | Source: 9618_s25_ms_13 (Q5.a.ii)

5(a)(ii)	1 mark for addressing mode, 1 mark for matching description • Direct • The operand is the address of the data • Indirect • The operand points to the memory location which is the address of the data • Indexed • The address of the data is formed by adding the contents of the Index Register (IX) to the operand	4
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Mark Scheme for Question 6 | Source: 9618_w24_ms_12 (Q8.a)

8(a)	1 mark for each correct content of the ACC (4) 1 mark for correct IX column	5																				
	<table><tr><td></td><td>Instructions</td><td>ACC content</td><td>IX content</td></tr><tr><td>1</td><td>LDM #19 DEC ACC</td><td>18</td><td>20</td></tr><tr><td>2</td><td>LDD 23 ADD 19</td><td>40</td><td>20</td></tr><tr><td>3</td><td>LDI 25 INC ACC</td><td>5</td><td>20</td></tr><tr><td>4</td><td>LDR #21 LDX 2</td><td>15</td><td>21</td></tr></table>		Instructions	ACC content	IX content	1	LDM #19 DEC ACC	18	20	2	LDD 23 ADD 19	40	20	3	LDI 25 INC ACC	5	20	4	LDR #21 LDX 2	15	21	
	Instructions	ACC content	IX content																			
1	LDM #19 DEC ACC	18	20																			
2	LDD 23 ADD 19	40	20																			
3	LDI 25 INC ACC	5	20																			
4	LDR #21 LDX 2	15	21																			

Mark Scheme for Question 7 | Source: 9618_w24_ms_13 (Q7.a)

7(a)	1 mark each:	3												
	<table border="1"> <thead> <tr> <th></th><th>Instructions</th><th>ACC content</th></tr> </thead> <tbody> <tr> <td>1</td><td>LDD 50 ADD #4 ADD 54</td><td>158</td></tr> <tr> <td>2</td><td>LDI 53 DEC ACC ADD 56</td><td>99</td></tr> <tr> <td>3</td><td>LDM #55 SUB #5</td><td>50</td></tr> </tbody> </table>		Instructions	ACC content	1	LDD 50 ADD #4 ADD 54	158	2	LDI 53 DEC ACC ADD 56	99	3	LDM #55 SUB #5	50	
	Instructions	ACC content												
1	LDD 50 ADD #4 ADD 54	158												
2	LDI 53 DEC ACC ADD 56	99												
3	LDM #55 SUB #5	50												

Mark Scheme for Question 8 | Source: 9618_w23_ms_11 (Q8.b.ii)

8(b)(ii)	1 mark for each bullet point (max 2) Similarity: • both load the contents of an <u>address</u> into the <u>Accumulator</u> Difference: • direct accesses the address given by the operand whereas indexed adds the contents of IX to the operand and accesses the data at that calculated address	2
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Mark Scheme for Question 9 | Source: 9618_w23_ms_11 (Q8.b.iii)

8(b)(iii)	1 mark for • Indirect (addressing) • Relative (addressing)	1
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Mark Scheme for Question 10 | Source: 9618_s23_ms_12 (Q3.b)

3(b)	1 mark for each correct term Direct addressing is when the operand holds the memory address of the data. Indirect addressing is when the operand holds a memory address that stores the memory address of the data. Immediate addressing is when the operand is the data.	3
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Mark Scheme for Question 11 | Source: 9618_w22_ms_13 (Q6.a.ii)

6(a)(ii)	1 mark for each bullet point: • To allow for re-locatable code • ... because all (target) addresses can be specified by the base address + offset	2
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Mark Scheme for Question 12 | Source: 9618_w21_ms_12 (Q8.b.iii)

8(b)(iii)	1 mark for name, 1 mark for description • Indirect addressing • the address to be used is at the given address • Relative addressing • the address to be used is an offset number of locations away, relative to the address of the current instruction • Indexed addressing • form the address from the given address plus the contents of the index register	2
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